Youssef Alaa Alfahl

AI/ML for Wireless Communications R&D Engineer

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Professional Summary

Communications and Information Engineering graduate with hands-on experience in wireless communication simulation, DSP, statistical analysis, and AI-based communication-system modeling. Built projects covering OFDM wireless links, convolutional channel coding, BPSK/AWGN transmission, Viterbi decoding, STFT-based spectrogram analysis, and large-scale data analysis. Experienced with Python, MATLAB, C/C++, signal processing, statistical reporting, and research documentation. Interested in R&D for low-power wireless systems, mesh networking, RF-constrained environments, and secure edge communication protocols.

Experience

University of Science and Technology in Zewail City — *[01/07/2024 – 30/9/2024]*

Sector: Communication and Information Engineering | **Role:** *Research Assistant*

Project: Deep Learning-Based RF Wave Modeling and Power Distribution Using Ray Tracing in Indoor Environments.

- **Simulation:** Generated indoor RF power maps in MATLAB using 3D modeling and ray tracing showing antenna patterns, reflections, materials.
- **Modeling:** Built power maps at user height and validated a U-Net to predict maps.
- **Outcome:** End-to-end pipeline for fast indoor coverage estimation across multiple scenarios.

National University of Sciences and Technology (NUST), Islamabad, Pakistan — *[08/07/2025 – 14/08/2025]*

Sector: School of Electrical Engineering and Computer Science (SEECS) | **Role:** Research Intern

Project: Multitask Deep Learning for Real-Time Optimization of Polarization Snapshots in Rotating-Polarization BPSK/QPSK

- **Simulation:** Modeled RP BPSK/QPSK links (AWGN/Rayleigh/Rician) and generated a Monte Carlo dataset in MATLAB.
- **Modeling:** Labeled optimal N_p under latency/ops-per-bit constraints and trained a multitask DL model to select N_p and predict BER.
- **Outcome:** Closely matched brute-force decisions with much lower computation, enabling real-time adaptive RP link configuration.

DHCB, Giza, Egypt — *[01/10/2025 – 1/12/2025]*

Sector: Software Development | **Role:** Full Stack Developer

Built and integrated software features using backend APIs and database-connected workflows, strengthening programming, debugging, documentation, and collaborative development skills.

University of Science and Technology in Zewail City — *[05/04/2026 – 14/06/2026]*

Sector: Education | **Role:** Part-Time Teaching Assistant | **Course:** CSCI 101 – Introduction to Computer Science (Python)

Support undergraduate students by teaching Python programming and guiding them through core problem-solving concepts.

Education

B.S.C., Communications and Information Engineering, University of Science and Technology in Zewail City

Enrollment: 03/10/2021 | **Graduation:** 01/02/2026 | **Final GPA:** 3.80 | **Relevant Coursework:** Wireless Communications, Digital Signal Processing, Communication Systems, Computer Networks, Probability & Stochastic Processes, Machine Learning, Neural Networks & Deep Learning, Embedded Systems.

Projects

OFDM Wireless Link Simulator (802.11a-Style) | Python

- Built an end-to-end OFDM wireless link simulator including FEC, interleaving, BPSK/QPSK/QAM mapping, subcarrier allocation, IFFT/FFT, cyclic prefix, and pilot tones.
- Modeled realistic transmitter/channel/receiver effects including PAPR reduction, Rapp HPA nonlinearity, FIR multipath channel, AWGN, and ZF/MMSE equalization.
- Evaluated link performance using BER vs SNR, EVM, constellation plots, PSD, spectrograms, and PAPR CCDF analysis.

JPEG Image Coding & Convolutional Channel Coding | Python

- Implemented a grayscale JPEG-style codec using custom DCT/IDCT, quantization, run-length coding, Huffman coding, and arithmetic coding.
- Designed a rate-1/3 convolutional encoder and hard-decision Viterbi decoder, integrated with BPSK modulation over an AWGN channel.
- Compared uncoded and channel-coded transmission using BER vs SNR curves and reconstructed image quality under different compression levels.

Spectrogram Analysis of Epileptic EEG Using STFT and Windowing | MATLAB

- Implemented STFT-based spectrogram generation from scratch using MATLAB FFT to analyze multi-channel epileptic EEG recordings.
- Studied the effect of window size, window type, and overlap ratio on time-frequency resolution using rectangular, triangular, Hamming, and Blackman windows.
- Built preprocessing and visualization scripts to compare ictal and seizure-free signal behavior across selected EEG channels.

COVID-19 Case Surveillance (CDC) | Python, DuckDB

- Built a scalable Python data analysis workflow to process and analyze a large CDC COVID-19 surveillance dataset with approximately 19 million rows.
- Performed validation, preprocessing, missing-data handling, descriptive analysis, hypothesis testing, and logistic regression.
- Produced visualizations and report-ready tables to communicate statistical findings, subgroup patterns, and outcome differences.

Technical Skills

Programming & Simulation: Python, MATLAB, C, C++, SQL

Wireless Communications & DSP: OFDM, BPSK, QPSK, QAM, AWGN, Rayleigh/Rician Channels, Multipath Channels, BER/SNR Analysis, EVM, PAPR, PSD, FFT/IFFT, STFT, Spectrogram Analysis

Channel Coding & Signal Processing: Convolutional Coding, Viterbi Decoding, Interleaving, DCT/IDCT, Quantization, Entropy Coding, Filtering, Time-Frequency Analysis

Data Analysis & ML/DL: Pandas, NumPy, SciPy, DuckDB, Parquet, Statistical Inference, Hypothesis Testing, Data Visualization Regression, Classification, Clustering, Neural Networks

Tools: Git, GitHub, Simulink, LabVIEW, Postman, Jupyter Notebook

Activities

- **Zewail City Science Festival 2022**: I contributed as a content creator and actor in scientific mini plays.
- **IEEE Zewail City Club**: I designed and conducted scientific experiments to illustrate key scientific concepts in different fields.